

3 Water and Life

KEY CONCEPTS

- 3.1** Polar covalent bonds in water molecules result in hydrogen bonding p. 45
- 3.2** Four emergent properties of water contribute to Earth's suitability for life p. 45
- 3.3** Acidic and basic conditions affect living organisms p. 51

Study Tip

Make a visual study guide: Draw a diagram and write a caption that explains how the structure of water supports life for each of the following properties of water:

The Properties of Water	
Cohesion of water molecules	Moderation of temperature
Floating of ice	The solvent of life

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For Students (in eText and Study Area)

- Get Ready for Chapter 3
- BioFlix® Animation: Adhesion and Cohesion in Plants
- Animation: Acids, Bases, and pH

For Instructors to Assign (in Item Library)

- Chemistry Review—Atoms and Molecules: Polar Attractions
- Chemistry Review—Water: Properties of Water

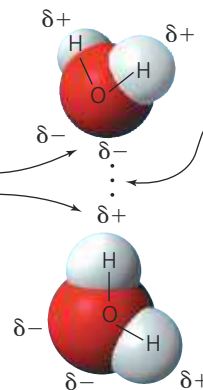


Figure 3.1 Ringed seals (*Phoca hispida*) depend on Arctic sea ice as a platform from which to hunt for fish in the water below. As Earth warms from climate change, the melting of sea ice is a threat to species that live on, under, and around the floating ice.

How does water's structure allow its solid form (ice) to float on liquid water?

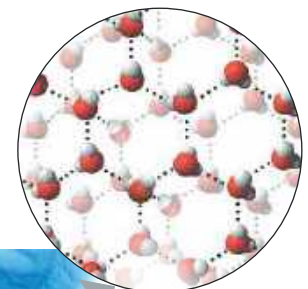
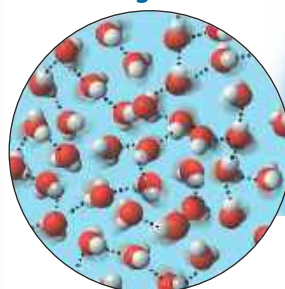
Water (H₂O) is a polar molecule:

At one end, the O has partial negative charges (δ^-) because O pulls electrons toward itself. At the other end, the H atoms have partial positive charges (δ^+).



Weak attractions between oppositely charged regions of water molecules, called **hydrogen bonds**, allow water molecules to bond to each other.

In liquid water, the hydrogen bonds constantly break and re-form. As a result, **the water molecules can slip closer together.**



In ice, the hydrogen bonds are stable and **the water molecules are farther apart.** Therefore, ice is less dense than liquid water, so it floats.

Floating ice insulates the water below, enabling survival of aquatic life. Water also has other life-supporting properties, as you'll see.